1 Elastic Launch Energy Worksheet

2 Definitions

Elastic potential energy (stored in a stretched rubber band or spring) is:

$$E_{\text{elastic}} = \frac{1}{2}kx^2 \tag{1}$$

where:

- E_{elastic} = elastic potential energy (J)
- k = spring constant (N/m)
- x = stretch distance (m)

Kinetic energy (KE) is the energy of motion:

$$E_k = \frac{1}{2}mv^2 \tag{2}$$

where:

- E_k = kinetic energy (J)
- m = mass of object (kg)
- v = velocity (m/s)

At a launch angle of 45°, the horizontal distance traveled is:

$$D = \frac{v^2}{g} \tag{3}$$

where:

- D = distance traveled (m)
- g = acceleration due to gravity (9.8 m/s^2)

3 Example Problems

Example 1: A rubber band with spring constant $k = 20$ N/m is stretched 0.15 m. Find the elastic potential energy stored.

$$\begin{aligned} E_{\text{elastic}} &= \frac{1}{2}(20)(0.15)^2 \\ &= \frac{1}{2}(20)(0.0225) \\ &= 0.225 \text{ J} \end{aligned}$$

Example 2: A 0.05 kg foam dart leaves the launcher at 6 m/s. Find its kinetic energy.

$$\begin{aligned} E_k &= \frac{1}{2}(0.05)(6)^2 \\ &= \frac{1}{2}(0.05)(36) \\ &= 0.9 \text{ J} \end{aligned}$$

Example 3: A dart is launched at 45° with velocity 8 m/s. Find the distance traveled.

$$\begin{aligned} D &= \frac{8^2}{9.8} \\ &= \frac{64}{9.8} \\ &\approx 6.53 \text{ m} \end{aligned}$$

4 Exercises

4.1 Elastic Energy

1. A spring with $k = 10$ N/m is stretched 0.2 m. Calculate E_{elastic}.
2. A rubber band launcher has $k = 25$ N/m and is stretched 0.1 m. How much energy is stored?
3. A spring with $k = 30$ N/m is stretched 0.05 m. Find the elastic potential energy.

4.2 Convert Elastic Energy to KE

Assume no energy loss ($E_{\text{elastic}} = E_k$)

1. A 0.1 kg object is launched using a spring stretched 0.2 m with $k = 15$ N/m. Find its kinetic energy.
2. **For a 4**: find its velocity.
3. A rubber band launcher has $k = 40$ N/m and stretch 0.1 m. It fires a 0.05 kg projectile. Find its kinetic energy.
4. **For a 4**: find its velocity.

4.3 Range at 45 Degrees

1. A 0.1 kg dart leaves at 5 m/s. How far does it travel?
2. A projectile launches at 7 m/s. Calculate the distance.
3. A 0.2 kg ball launches at 10 m/s. How far does it travel?
4. A projectile has speed 9 m/s. If $g = 9.8$, how far will it go?
5. **For a 4**: A foam ball travels 6 m when fired at 45°. Solve for v.
6. **For a 4**: A launcher fires a ball 8 m. Find the speed needed.